Prime Projections on Zero-Dependent Exponential Lattices

Abstract

We associate to each prime number p_k a nearby real number v_k obtained by projecting p_k onto an exponential lattice whose ratio is determined by the ordinate γ_k of a nontrivial zero of the Riemann zeta function. Under the standing assumption that the nontrivial zeros lie on the critical line, the positive ordinates may be ordered as a sequence (γ_k), counted with multiplicity, and the Riemann–von Mangoldt formula gives their first-order asymptotic distribution.

For each k, the integer n_k denotes the first lattice index for which the corresponding lattice point lies above p_k, and v_k denotes that lattice point. The construction gives

$$p_k \le v_k \le p_k \exp\left(\frac{2\pi}{\gamma_k}\right),$$

and therefore

$$\frac{v_k}{p_k} \to 1.$$

Writing

$$d_k := v_k - p_k$$

for the additive displacement, the Riemann–von Mangoldt formula and the Prime Number Theorem imply

$$d_k = O\big((\log k)^2\big).$$

The displacement is governed by an exact phase factor

$$\eta_k = n_k - \frac{\gamma_k \log p_k}{2\pi}, \qquad 0 \le \eta_k < 1.$$

This phase gives the exact identity

$$v_k = p_k \exp\left(\frac{2\pi\eta_k}{\gamma_k}\right),$$

and hence an exact expression for d_k. A refined upper envelope for d_k is obtained by combining the asymptotic inversion of the zero-counting formula with Cipolla's expansion for the k-th prime. The counting function

$$\pi_V(x) := \#\{k : v_k \le x\}$$

satisfies

$$\pi_V(x) \sim \pi(x) \sim \mathrm{Li}(x).$$

Numerical computations show non-monotone behaviour of (v_k), and the descent criterion is expressed explicitly in terms of the phase-dependent displacements.

1 Introduction

Let $(p_k)_{k\ge1}$ denote the sequence of prime numbers in increasing order. Let $(\gamma_k)_{k\ge1}$ denote a sequence of positive ordinates associated with the nontrivial zeros of the Riemann zeta function.

Throughout this article we work under the following standing assumption.

Standing assumption. Every nontrivial zero of $\zeta(s)$ has the form

$$\rho = \frac{1}{2} + i\gamma.$$

The positive ordinates are ordered increasingly and counted with multiplicity:

$$0 < \gamma_1 \le \gamma_2 \le \gamma_3 \le \cdots .$$

Let

$$N(T) := \#\{0 < \gamma \le T : \zeta(1/2 + i\gamma) = 0\},$$

where zeros are counted with multiplicity. The Riemann–von Mangoldt formula gives [2–4]

$$N(T) = \frac{T}{2\pi}\log\left(\frac{T}{2\pi}\right) - \frac{T}{2\pi} + O(\log T). \tag{1}$$

Inverting (1) gives

$$\gamma_k \sim \frac{2\pi k}{\log k}. \tag{2}$$

We shall also use the Prime Number Theorem in the form [5–8]

$$p_k \sim k \log k. \tag{3}$$

The aim of this article is to study the sequence obtained by moving each prime p_k to the next point of the exponential lattice

$$\left\{\exp\left(\frac{2\pi n}{\gamma_k}\right) : n \in \mathbb{Z}\right\}.$$

Since the ratio between consecutive lattice points is

$$\exp\left(\frac{2\pi}{\gamma_k}\right),$$

and since $\gamma_k \to \infty$, the projection is multiplicatively small.

The sequence produced in this way satisfies

$$v_k \sim p_k.$$

Moreover, using (1) and (3), its additive displacement satisfies

$$v_k - p_k = O\big((\log k)^2\big).$$

The leading scale of the upper envelope can be refined further by combining Cipolla's expansion for the k-th prime with the asymptotic inversion of (1) [1–3, 8].

A central role is played by the exact phase decomposition

$$\frac{\gamma_k \log p_k}{2\pi} = n_k - \eta_k, \qquad 0 \le \eta_k < 1,$$

which rewrites the zero ordinate relative to the logarithmic scale $\log p_k$. This decomposition is an identity, not an independent approximation of γ_k. Its role is structural: it separates the integer lattice coordinate n_k from the bounded phase factor η_k, and thereby gives an exact formula for the local displacement $v_k - p_k$.

Numerical computations show non-monotonicity of (v_k). The descent condition is determined by the competition between prime gaps and the variation of the phase-dependent displacements.

2 Notation

Throughout the paper we use the following notation.

- p_k denotes the k-th prime number.
- γ_k denotes the k-th positive ordinate of a nontrivial zero of $\zeta(s)$, counted with multiplicity.
- n_k denotes the integer lattice index associated with p_k and γ_k.
- v_k denotes the projected lattice point associated with p_k and γ_k.
- $d_k := v_k - p_k$ denotes the additive displacement.
- η_k denotes the phase factor
$$\eta_k := n_k - \frac{\gamma_k \log p_k}{2\pi}.$$
- $\pi(x)$ denotes the prime counting function.
- $\pi_V(x)$ denotes the counting function
$$\pi_V(x) := \#\{k : v_k \le x\}.$$
- $\mathrm{Li}(x)$ denotes the logarithmic integral.

We use the standard asymptotic notation $f(x) \sim g(x)$ to mean

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 1.$$

We write $f(x) = O(g(x))$ if there exists a constant $C > 0$ such that

$$|f(x)| \le C|g(x)|$$

for all sufficiently large x.

3 Definition of the projected sequence

For each $k \ge 1$, consider the exponential lattice

$$\mathcal{L}_k := \left\{ \exp\left(\frac{2\pi n}{\gamma_k}\right) : n \in \mathbb{Z} \right\}.$$

Equivalently, $\mathcal{L}_k$ is the set of positive real numbers x satisfying

$$\gamma_k \log x \in 2\pi\mathbb{Z}. \tag{4}$$

Writing

$$x_{k,n} := \exp\left(\frac{2\pi n}{\gamma_k}\right), \qquad n \in \mathbb{Z}, \tag{5}$$

we have

$$\frac{x_{k,n+1}}{x_{k,n}} = \exp\left(\frac{2\pi}{\gamma_k}\right). \tag{6}$$

Definition 3.1. For each $k \geq 1$, define the integer

$$n_k := \left\lceil \frac{\gamma_k \log p_k}{2\pi} \right\rceil. \tag{7}$$

The projected value v_k is then defined by

$$v_k := x_{k,n_k} = \exp\left(\frac{2\pi n_k}{\gamma_k} \right). \tag{8}$$

Equivalently, v_k is the first point of $\mathcal{L}_k$ lying above p_k.

The integer n_k is characterized by

$$x_{k,n_k-1} < p_k \leq x_{k,n_k}, \tag{9}$$

except when $p_k = x_{k,n_k}$, in which case the projected point is exactly p_k. The estimates below are unaffected by this boundary case.

Lemma 3.2. *For every $k \geq 1$,*

$$v_k \geq p_k.$$

Proof. By Definition 3.1, n_k is chosen so that $x_{k,n_k} \geq p_k$. Since $v_k = x_{k,n_k}$, the result follows. □

Lemma 3.3. *For every $k \geq 1$,*

$$p_k \leq v_k \leq p_k \exp\left(\frac{2\pi}{\gamma_k} \right). \tag{10}$$

Proof. By Lemma 3.2, $v_k \geq p_k$. Moreover, by (7),

$$n_k = \left\lceil \frac{\gamma_k \log p_k}{2\pi} \right\rceil \leq \frac{\gamma_k \log p_k}{2\pi} + 1.$$

Therefore

$$v_k = \exp\left(\frac{2\pi n_k}{\gamma_k} \right) \leq \exp\left(\log p_k + \frac{2\pi}{\gamma_k} \right) = p_k \exp\left(\frac{2\pi}{\gamma_k} \right).$$

This proves (10). □

4 Exact phase representation

The definition of n_k gives an exact decomposition of the quantity

$$\frac{\gamma_k \log p_k}{2\pi}$$

into an integer part and a bounded phase. Define

$$\eta_k := n_k - \frac{\gamma_k \log p_k}{2\pi}. \tag{11}$$

Since n_k is the ceiling in (7), we have

$$0 \leq \eta_k < 1. \tag{12}$$

Rearranging (11) gives the exact identity

$$\gamma_k = \frac{2\pi(n_k - \eta_k)}{\log p_k}. \tag{13}$$

Thus (n_k, η_k) is an exact phase decomposition of the ordinate γ_k relative to the logarithmic scale $\log p_k$.

Substituting

$$n_k = \frac{\gamma_k \log p_k}{2\pi} + \eta_k$$

into (8), we obtain the exact formula

$$v_k = p_k \exp\left(\frac{2\pi\eta_k}{\gamma_k}\right). \tag{14}$$

Consequently, the displacement $d_k = v_k - p_k$ is exactly

$$d_k = p_k \left[\exp\left(\frac{2\pi\eta_k}{\gamma_k}\right) - 1\right]. \tag{15}$$

If the bounded phase is discarded from (13), one obtains the integer approximation

$$\gamma_k = \frac{2\pi n_k}{\log p_k} + O\left(\frac{1}{\log p_k}\right). \tag{16}$$

More precisely,

$$\gamma_k - \frac{2\pi n_k}{\log p_k} = -\frac{2\pi\eta_k}{\log p_k}, \tag{17}$$

and therefore

$$\left|\gamma_k - \frac{2\pi n_k}{\log p_k}\right| < \frac{2\pi}{\log p_k}. \tag{18}$$

Remark 4.1. The identity (13) is not an independent approximation to γ_k, because both n_k and η_k are defined from γ_k. Its role is to provide an exact coordinate system: an integer lattice coordinate n_k and a bounded phase η_k.

5 The displacement

The exact displacement formula (15) immediately yields the basic estimates.

Lemma 5.1. *For every $k \geq 1$,*

$$0 \leq d_k \leq p_k \left(\exp\left(\frac{2\pi}{\gamma_k}\right) - 1\right). \tag{19}$$

Consequently,

$$d_k = O\left(\frac{p_k}{\gamma_k}\right). \tag{20}$$

Proof. Since $0 \leq \eta_k < 1$, (15) gives

$$0 \leq d_k \leq p_k \left(\exp\left(\frac{2\pi}{\gamma_k}\right) - 1\right).$$

As $\gamma_k \to \infty$,

$$\exp\left(\frac{2\pi}{\gamma_k}\right) - 1 = \frac{2\pi}{\gamma_k} + O\left(\frac{1}{\gamma_k^2}\right),$$

and therefore

$$d_k = O\left(\frac{p_k}{\gamma_k}\right).$$

□

Proposition 5.2. *One has*

$$d_k = O\big((\log k)^2\big). \tag{21}$$

Proof. By Lemma 5.1,

$$d_k = O\left(\frac{p_k}{\gamma_k}\right).$$

Using (3) and (2), we get

$$\frac{p_k}{\gamma_k} \sim \frac{k \log k}{2\pi k/\log k} = \frac{(\log k)^2}{2\pi}.$$

Hence

$$d_k = O\big((\log k)^2\big).$$

□

Remark 5.3. Proposition 5.2 is an upper bound. It does not imply

$$d_k \sim (\log k)^2.$$

The actual displacement depends on the bounded phase η_k, as shown in (15).

5.1 Refined upper envelope

The estimate (21) can be refined at the level of the upper envelope. Let

$$L_k := \log p_k, \qquad L_{2,k} := \log\log p_k. \tag{22}$$

Combining the asymptotic inversion of (1) with Cipolla's expansion for the k-th prime gives, for every fixed $J \geq 1$, an expansion of the form [1–3, 8]

$$\gamma_k = \frac{2\pi p_k}{L_k^2}\left(1 + \sum_{j=1}^{J} \frac{P_j(L_{2,k})}{L_k^j} + O\left(\frac{L_{2,k}^{J+1}}{L_k^{J+1}}\right)\right), \tag{23}$$

where each P_j is a real polynomial of degree at most j. The precise coefficients are not needed in what follows.

Inverting (23) gives

$$\frac{2\pi p_k}{\gamma_k} = L_k^2\left(1 + \sum_{j=1}^{J} \frac{Q_j(L_{2,k})}{L_k^j} + O\left(\frac{L_{2,k}^{J+1}}{L_k^{J+1}}\right)\right), \tag{24}$$

where each Q_j is again a real polynomial of degree at most j. In particular,

$$\frac{2\pi p_k}{\gamma_k} \sim (\log p_k)^2. \tag{25}$$

Define the local upper envelope

$$A_k := p_k\left(\exp\left(\frac{2\pi}{\gamma_k}\right) - 1\right). \tag{26}$$

By Lemma 5.1,

$$0 \leq d_k \leq A_k. \tag{27}$$

Since

$$\exp u - 1 = u + O(u^2) \qquad (u \to 0),$$

we have

$$A_k = \frac{2\pi p_k}{\gamma_k} + O\left(\frac{p_k}{\gamma_k^2}\right). \tag{28}$$

Using (23),

$$\frac{p_k}{\gamma_k^2} = O\left(\frac{(\log p_k)^4}{p_k}\right) = o(1). \tag{29}$$

Thus

$$A_k = \frac{2\pi p_k}{\gamma_k} + o(1). \tag{30}$$

Combining (24) and (30), we obtain

$$A_k = L_k^2 \left(1 + \sum_{j=1}^{J} \frac{Q_j(L_{2,k})}{L_k^j} + O\left(\frac{L_{2,k}^{J+1}}{L_k^{J+1}}\right)\right) + o(1). \tag{31}$$

Therefore,

$$0 \le d_k \le L_k^2 \left(1 + \sum_{j=1}^{J} \frac{Q_j(L_{2,k})}{L_k^j} + O\left(\frac{L_{2,k}^{J+1}}{L_k^{J+1}}\right)\right) + o(1). \tag{32}$$

The actual displacement is obtained from this envelope through the phase factor η_k. Indeed, expanding (15) gives

$$d_k = \eta_k \frac{2\pi p_k}{\gamma_k} + O\left(\frac{p_k}{\gamma_k^2}\right), \tag{33}$$

and hence, by (29),

$$d_k = \eta_k \frac{2\pi p_k}{\gamma_k} + o(1). \tag{34}$$

Combining (34) with (24) gives

$$d_k = \eta_k L_k^2 \left(1 + \sum_{j=1}^{J} \frac{Q_j(L_{2,k})}{L_k^j} + O\left(\frac{L_{2,k}^{J+1}}{L_k^{J+1}}\right)\right) + o(1). \tag{35}$$

6 Asymptotic equivalence

Theorem 6.1. *One has*

$$\frac{v_k}{p_k} \to 1 \qquad (k \to \infty). \tag{36}$$

Proof. By Lemma 3.3,

$$1 \le \frac{v_k}{p_k} \le \exp\left(\frac{2\pi}{\gamma_k}\right).$$

Since $\gamma_k \to \infty$,

$$\exp\left(\frac{2\pi}{\gamma_k}\right) \to 1.$$

The result follows by squeezing. □

7 The counting function

Define

$$\pi_V(x) := \#\{k : v_k \le x\}. \tag{37}$$

Lemma 7.1. *For every $x > 0$,*

$$\pi_V(x) \le \pi(x). \tag{38}$$

Proof. If $v_k \le x$, then $p_k \le v_k \le x$. Therefore every index counted by $\pi_V(x)$ is also counted by $\pi(x)$. □

Lemma 7.2. *For every $\delta > 0$, there exists $K_\delta \ge 1$ such that, for all sufficiently large x,*

$$\pi\left(\frac{x}{1+\delta}\right) - K_\delta \le \pi_V(x). \tag{39}$$

Proof. By Theorem 6.1, for every $\delta > 0$ there exists K_δ such that

$$v_k \le (1+\delta)p_k \tag{40}$$

for all $k \ge K_\delta$. If

$$p_k \le \frac{x}{1+\delta}$$

and $k \ge K_\delta$, then

$$v_k \le (1+\delta)p_k \le x.$$

Thus all sufficiently large primes below $x/(1+\delta)$ contribute to $\pi_V(x)$, proving (39). □

Theorem 7.3. *The counting function $\pi_V(x)$ satisfies*

$$\pi_V(x) \sim \pi(x). \tag{41}$$

Consequently,

$$\pi_V(x) \sim \mathrm{Li}(x). \tag{42}$$

Proof. From Lemmas 7.1 and 7.2, for every fixed $\delta > 0$,

$$\pi\left(\frac{x}{1+\delta}\right) - K_\delta \le \pi_V(x) \le \pi(x). \tag{43}$$

Dividing by $\pi(x)$ and using the Prime Number Theorem [5–8], we get

$$\frac{\pi(x/(1+\delta))}{\pi(x)} \to \frac{1}{1+\delta}.$$

Since $K_\delta/\pi(x) \to 0$,

$$\frac{1}{1+\delta} \le \liminf_{x\to\infty} \frac{\pi_V(x)}{\pi(x)} \le \limsup_{x\to\infty} \frac{\pi_V(x)}{\pi(x)} \le 1.$$

Letting $\delta \to 0$ proves (41). Since

$$\pi(x) \sim \mathrm{Li}(x),$$

we also obtain (42). □

Remark 7.4. The proof of Theorem 7.3 does not require monotonicity of the sequence (v_k). The counting function $\pi_V(x)$ counts all indices k satisfying $v_k \le x$, independently of the order of the real numbers v_k.

8 Non-monotonicity

The sequence (v_k) satisfies

$$v_{k+1} - v_k = (p_{k+1} - p_k) + (d_{k+1} - d_k). \tag{44}$$

Thus a descent

$$v_{k+1} < v_k$$

occurs precisely when

$$d_k - d_{k+1} > p_{k+1} - p_k. \tag{45}$$

Using the phase formula (34), the displacement difference is governed by the successive phase factors η_k and η_{k+1}, together with the slowly varying envelope $2\pi p_k/\gamma_k$.

Numerical computations show that the condition (45) occurs, and hence that the sequence (v_k) is not monotone. This behaviour is consistent with the fact that the upper envelope of d_k has order $(\log k)^2$, whereas the average prime gap has order $\log k$. However, the actual occurrence of descents is controlled by the phase factors η_k, not by the envelope alone.

Proposition 8.1. *If there exist infinitely many indices k for which*

$$d_k - d_{k+1} > p_{k+1} - p_k,$$

then (v_k) has infinitely many descents.

Proof. For every such k, equation (44) gives

$$v_{k+1} - v_k = (p_{k+1} - p_k) - (d_k - d_{k+1}) < 0.$$

Thus $v_{k+1} < v_k$. If the inequality occurs for infinitely many k, then there are infinitely many descents. □

Remark 8.2. The numerical non-monotonicity of (v_k) is naturally interpreted through the exact descent condition (45). The estimate

$$d_k = O\big((\log k)^2\big)$$

gives the available scale, while the exact phase formula (15) determines whether a particular step produces an ascent or a descent.

9 Concluding remarks

We have constructed a real sequence (v_k) by projecting each prime p_k onto the next point of the exponential lattice

$$\mathcal{L}_k = \left\{ \exp\left(\frac{2\pi n}{\gamma_k}\right) : n \in \mathbb{Z} \right\}.$$

The integer lattice index is

$$n_k = \left\lceil \frac{\gamma_k \log p_k}{2\pi} \right\rceil,$$

and the projected value is

$$v_k = \exp\left(\frac{2\pi n_k}{\gamma_k}\right).$$

The basic estimate

$$p_k \le v_k \le p_k \exp\left(\frac{2\pi}{\gamma_k}\right)$$

implies

$$v_k \sim p_k.$$

The associated counting function satisfies

$$\pi_V(x) \sim \pi(x) \sim \mathrm{Li}(x).$$

The exact phase representation

$$\eta_k = n_k - \frac{\gamma_k \log p_k}{2\pi}, \qquad 0 \le \eta_k < 1,$$

gives

$$\gamma_k = \frac{2\pi(n_k - \eta_k)}{\log p_k}$$

and

$$v_k = p_k \exp\left(\frac{2\pi\eta_k}{\gamma_k}\right).$$

Thus the displacement

$$d_k := v_k - p_k$$

is exactly

$$d_k = p_k \left[\exp\left(\frac{2\pi\eta_k}{\gamma_k}\right) - 1\right].$$

The displacement satisfies

$$d_k = O\big((\log k)^2\big),$$

and its upper envelope admits a refined asymptotic expansion obtained from Cipolla's expansion and the inverted zero-counting formula. The actual displacement is obtained from this envelope by multiplication, to first order, by the phase factor η_k.

The sequence is globally prime-like in first-order density, while its local behaviour is governed by the interaction between prime gaps and the phase-dependent displacements. Numerical computations show non-monotonicity, and the descent criterion

$$d_k - d_{k+1} > p_{k+1} - p_k$$

identifies the precise mechanism by which descents occur.

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